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Scaleform®

Scaleform 4.3 .

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: 3.01

: 2011 9 15

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1 환영인사

CRI™ Scaleform Video Scaleform™ .

Scaleform Video

, .

, , HUD, , (cut scene)

.

CRI . CRI

,

. CRI 1700 .

Scaleform Video PC (X360, PS3, Wii) , ,

, .

. Scaleform Video

, , , , ,

.

Scaleform Video ,

, ,

(export) . ,

5.1 ,

. Scaleform Video ,

, (cut scene) (cue point)

.

Scaleform Video SDK :

- Scaleform Video
- Scaleform Video
- /
- Scaleform Video ( )

2 스케일폼 비디오에 관하여

, .

.

2.1

:

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder

:

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Video

3

.

. USM .

Scaleform videodemo.swf .

2.2

,

.

- *getting_started_with_video_tutorial.fla –*

- *getting_started_with_video_tutorial.swf –*
- *sample.avi –*
- *sample_audio.wav –*
- *sample_cue_points.txt –*
- *sample_subtitles.txt –*

2.3

.

- Medianoche.exe —
- ScaleformVideoEncoder.exe — ( )
- TMPGLib.dll —
- VideoEncoderUtil.dll —ScaleformVideoEncoder.exe
- VideoPlayer.swf — (Preview)

2.4

- **고급 무비 재생**

  API . HD(1080p)

  .

- **다중언어 및 자막 지원**

  . ( )

  .

- **쉬운 조작과 전문가 수준 인코딩**

  SDK .

2.5

CRI Movie

. , , ,

. , ,

3D , .

, , ,

HUD, , , , .

Scaleform 4.1

. CRI Movie

.

2.6

- **비디오와 오디오 포맷:** , AVI WAV

- **멀티 플랫폼 지원:** XB360, PS3, PC, Wii

- **다른 플랫폼용 최적화 인코딩:**

/

- **자막:** .

,

. Scaleform

.

- **알파채널 지원:** " "

, .

.

- **텍스처에 렌더:**

.

- **오디오 트랙:** (32) .

.

- **오디오 트랙 전환:** Scaleform 4.3

.

- **지역화 지원:**

  .

  UI .

- **서라운드 사운드 지원:** / /5.1

  .

- **다중언어 서라운드 사운드:** ( )

  . .

- **탐색 능력:** Scaleform Video .

  .

  (rewind), (forward) .

- **큐포인트:** .

  .

  USM .

  .

  ( ) .

  ,

  .

2.7

CRI USM . USM

Scaleform Media Player .

.

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Video\scaleform_logo.usm

.

2.8

그림 1: 스케일폼 비디오 워크플로우

.

1. API
2. AVI
3. AVI USM
4. USM SWF
5. , ,
6. SWF
7. SWF

주의: USM , .

. .

3 시작하기: 스케일폼 비디오 인코딩

Scaleform 4.3 CRI USM . CRI

HD .

. USM , ,

.

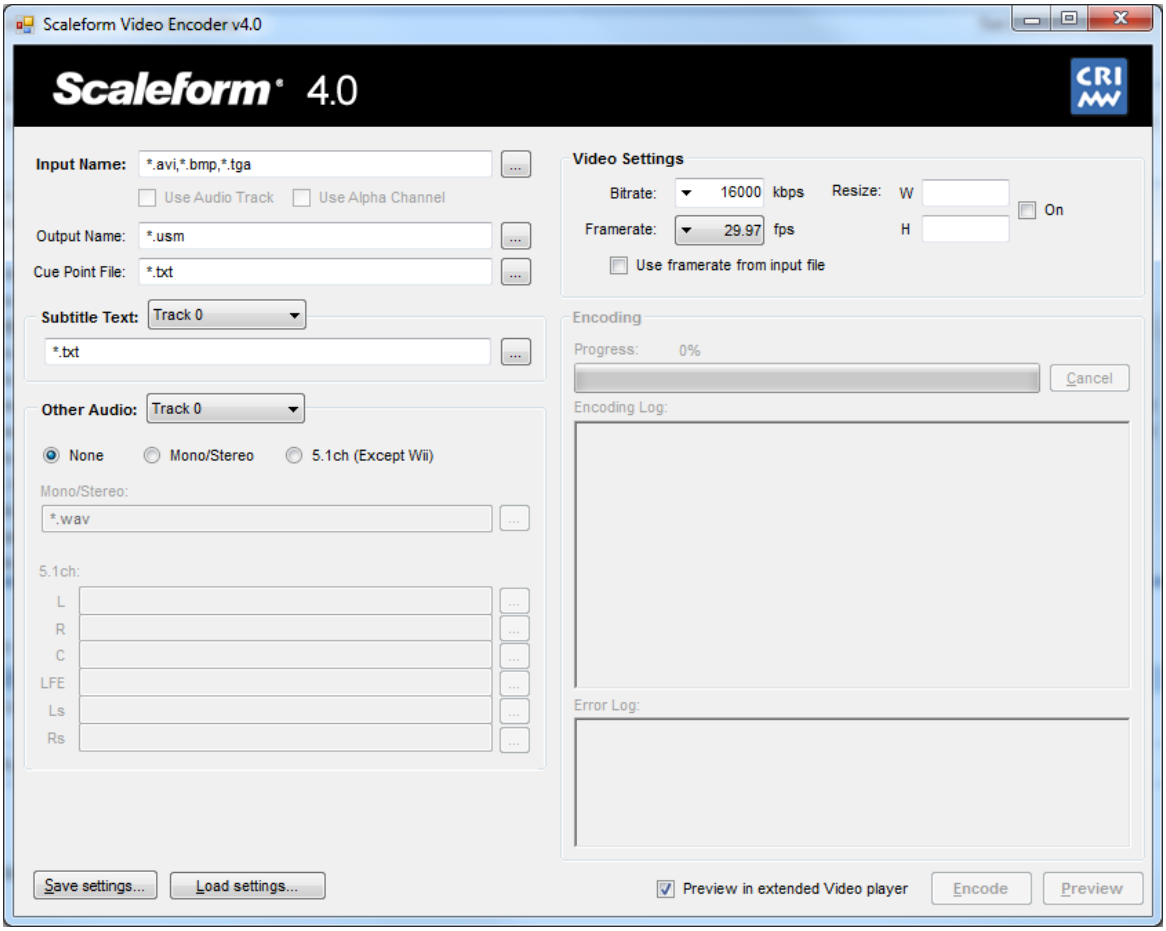


그림 2: 스케일폼 비디오 인코더

3.1 -

USM . ,

. ( ), , , (2)

USM .

.

.

Windows Vista: -> *->Scaleform->GFx SDK 4.3->Video*

Windows XP: -> *->Scaleform->GFx SDK 4.3->Video*

1. Browse AVI .
   a. AVI BMP TGA
   b. BMP TGA .

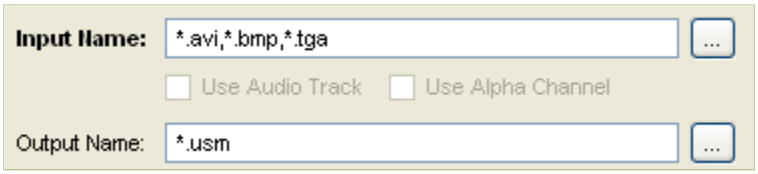


그림 3: 인코딩 될 비디오 파일은 Input Name 항목에 들어가야 함

2. *Use Audio Track* .
   AVI , BMP TGA .

3. *Use Alpha Channel* .

. .

4. *Output Name* . , *browse*
   USM .

5. *Encode* .

3.2

1.

2. *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\Tools\VideoEncoder\sample.avi* .

3. .

4. , Preview . USM preview .

3.3

, , SWF .

, .

USM Cue Point File Browse

.

그림 4: 큐 포인트 텍스트 파일 탐색

.

(1000) .

. 5000 5 (5000/1000=5).

.

2 , .

DVD .

. " = " .

, (,) .

" " .

한줄 큐포인트 포맷

```
time, cue point type (0 or 1), cue point name, parameter1, parameter2, …
paramter10
```

사용예

```
1000, 0, cue_point_1, my_param=5, my_other_param=10
```

예제 CutPoint.txt 파일의 내용

```
1000
0,0,start
1000,0,cue1
2000,1,cue2,name1_0=value1_0,name1_1=value1_1
3000,0,cue3
4000,1,cue4,name4_0=value4_0
5000,0,cue5
6000,1,cue6,name6_0=value6_0,name6_1=value6_1
7000,0,cue7,name7_0=value7_0,name7_1=value7_1,name7_2=value7_2
8000,1,cue8,name8_0=value_0
9000,0,cue9
10000,1,cue10,name10_0=value10_0,name10_1=value10_1
11000,0,end
```

3.3.1 샘플 비디오를 큐포인트로 재인코딩 하기

1.

2. *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder \sample.avi*

3. Cue Point File *sample_cue_points.txt* (
Cueint

3.4.1 자막으로 샘플 비디오 재인코딩

1.

2. *C:\Program Files\Scaleform\GFx SDK 4.3 \Bin\Tools\VideoEncoder\sample.avi*

3. *Cue Point File* browse sample_cue_points.txt (*sample.avi*)

4. *Subtitle Text* browse sample_subtitles.txt (*sample.avi*)

5.

6. (Scaleform Media Player) Scaleform Player *getting_started_with_video_tutorial.swf* .

7. . 2-5 8-11

5.1 :

L –

R –

C – , ( )

LFE –

Ls –

Rs –

3.5.1 두개의 트랙으로 예제 비디오를 재인코딩

1.
2. *C:\Program Files\Scaleform\GFx SDK 4.3 \Bin\Tools\VideoEncoder\sample.avi*
3. *Cue Point File* *browse* *sample_cue_points.txt* - *sample.avi*
4. browse sample_subtitles.txt - sample.avi
5. *Other Audio* Track 1
6. Mono/Stereo
7. *sampe_audio.wav*
8. . ,
   , .
9. *Preview* USM
10. *Audio Channel* ,
    .

3.6

, , .

(, , ,)

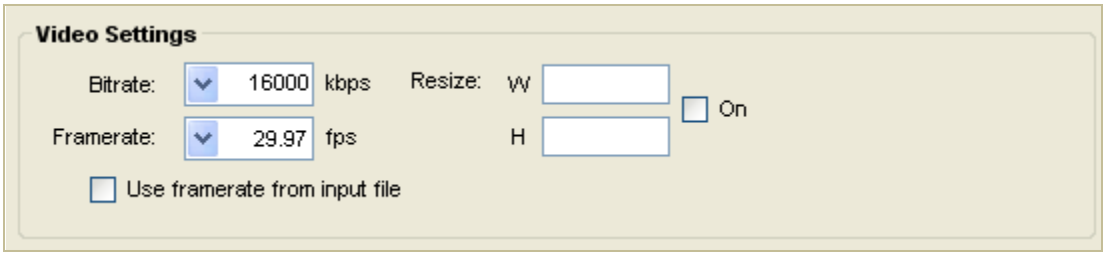


그림 7: 비디오 설정

1. — . .

. ,

. , ,

.

.

.

.

(kbps)					
Video Format		**PS3**	**Xbox360**	**Wii**	**PC**
1080p	High Quality	40000	40000	n/a	40000
1080p	Standard	30000	30000	n/a	30000
1080p	High Compression	20000	20000	n/a	20000
720p	High Quality	36000	36000	n/a	36000
720p	Standard	26000	26000	n/a	26000
720p	High Compression	16000	16000	n/a	16000
480p	High Quality	16000	16000	8000	16000
480p	Standard	12000	12000	6000	12000
480p	High Compression	8000	8000	4000	8000

2. ( ) . 29.97 fps(NTSC
) . AVI *Use framerate from input file*
, .
. 1
. NTSC 1
29.97 . PC
60 60 .
3. ( ) *Resize W* and *H* . W H
.
4. , .
USM .
.
5. *Preview* .
Encode *Preview in extended Video player option*
preview .

4 시작하기: 비디오를 플래시에 추가

, USM

. SWF .

play, pause, rewind,

. **이에 대한 상세한 방법은 NetStream 클래스에 관한 플래시 대한**

4.1

AS 2.0

. USM SWF

.

1. Layer 1 'mvplayer' .

2. *New Layer* 'actions' .

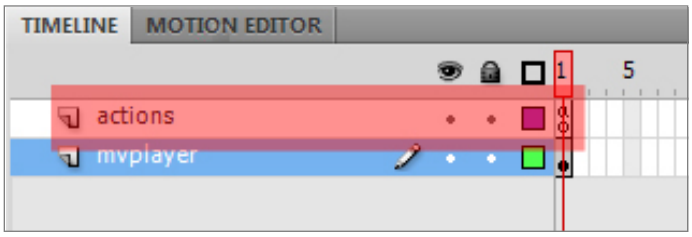


그림 8: 타임라인의 액션과 mvplayer 레이어

3. *actions* 1 F9 *Actions* .

.

```
_global.gfxExtensions = true;
_focusrect = false;
onLoad = function()
{
   mvplayer.playVideo("sample.usm");
}
```

주의: SWF USM

. ,

, . - :

"C:/pathtovideo/sample.usm"

4. mvplayer

5.

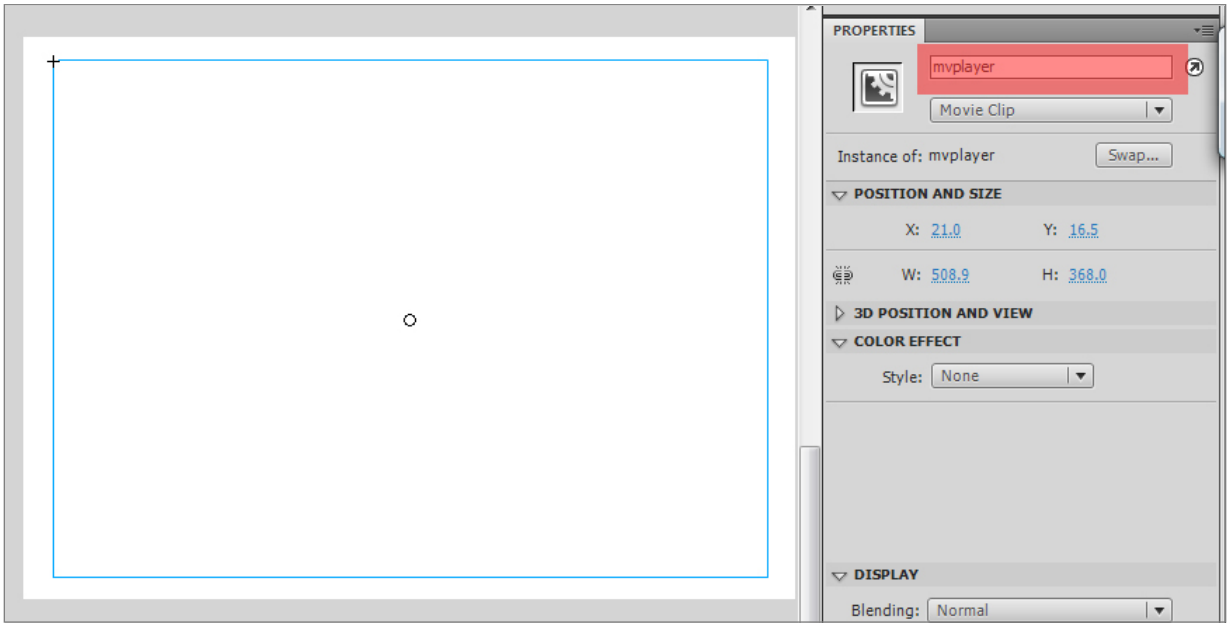
a. ,

.

b.

c. *Convert to Symbol*

d. 'mvplayer' OK



S

그림 9: 무비클립 만들기, mvplayer

6. mvplayer

7. *Properties* 'mvplayer'

8. mvplayer .

9. *mvplayer* 'actions' .

10. *actions* 1 *Actions*

```
nc = new NetConnection();
nc.connect(null);
ns = new NetStream(nc);
video.attachVideo(ns);
this.attachAudio(ns);
var audio_sound:Sound = new Sound(this);
function playVideo(videoToPlay:String):Void
{
      ns.play(videoToPlay);
}
```

주의: mvplayer 1 actions ,

Layer 1

11. *Library* *New Video*

12. 'video' OK

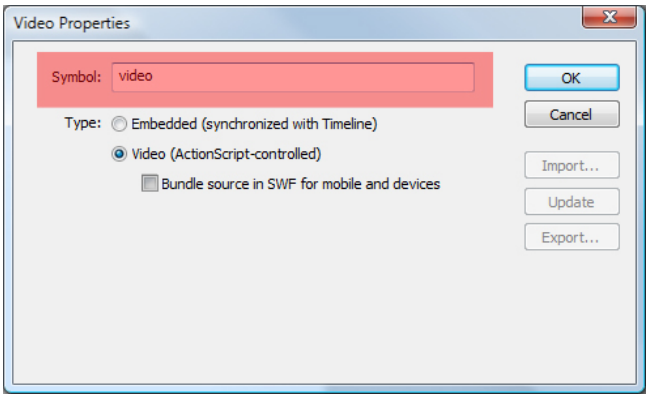


그림 10: 비디오 객체 만들기

13. *New Layer* . 'Layer 3'

14. *Library* new video *Stage* *Layer3* 1 .

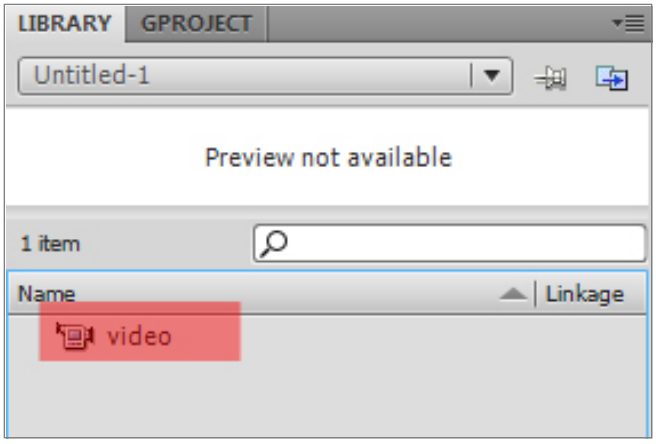


그림 11: 라이브러리 창의 비디오 객체

15. *Properties* 'video'

16.

4.2

SWF Scaleform Player

.

1. SWF

:

1. Scaleform Player

2. Scaleform Player SWF

   a. SWF Scaleform Player

5 시작하기: 액션스크립트에서 비디오 작업하기

USM(, ,) ,

, , ,

. : " : "

.

SWF , , .

.

.

5.1

USM .

,

.

주의: USM SWF

.

USM .

" " " " .

. mvplayer

actions 1 .

.

1. *mvplayer* .
2. *mvplayer* actions

   Actions .
3. traceCuePoint .

```
ns.onCuePoint = traceCuePoint;
```

4. traceCuePoint . ,

   metaPropPJParams . ,

   ( ) . (currentCuePoint.name),

   (currentCuePoint.time), (currentCuePoint.type) .

   (ns.time) .

```
function traceCuePoint(currentCuePoint:Object):Void
{
      var metaPropPJParams:Object = currentCuePoint.parameters;
```

```
        trace("\t\t name: " + currentCuePoint.name);
        trace("\t\t time: " + currentCuePoint.time);
        trace("\t\t type: " + currentCuePoint.type);
        if (metaPropPJParams != undefined)
            {
            trace("\t\t parameters:");
            traceObject(metaPropPJParams, 4);
        }
    }
```

5. , traceObject

. SWF ( )

.

```
function traceObject(obj:Object, indent:Number):Void
{
      var indentString:String = "";
      for (var j:Number = 0; j < indent; j++)
      {
      indentString += "\t";
      }
      for (var i:String in obj)
      {
      if (typeof(obj[i]) == "object")
            {
            trace(indentString + " " + i + ": [Object]");
            traceObject(obj[i], indent + 1);
         }
            else
            {
            trace(indentString + " " + i + ": " + obj[i]);
            }
      }
}
```

6. SWF Scaleform Player

. Scaleform Player

.

5.2

.

.

주의: USM SWF

. .

" " " " .

1. *mvplayer*

2. *mvplayer* actions . *Actions*

3. , subtitleChannelNumber 0

```
var subtitleChannelNumber = 0;
```

4. , . : 0

.

```
ns.onMetaData = function(infoObject:Object)
{
      ns.subtitleTrack = 1;
      subtitleChannelNumber = infoObject.subtitleTracksNumber;
       trace("Subtitle Channels: " +  subtitleChannelNumber);
}
```

5. *mvplayer*

'subtitle' . .

6. . msg

USM .

subtitleId2

. Msg 5 *subtitle* .

```
ns.onSubtitle = function(msg:String)
{
   trace("Subtitle: " + msg);
   subtitle.text = msg;
}
```

7. SWF Scaleform Player

.

.

5.3

USM

. SWF

.

USM

SWF . "

" "

" .

mvplayer *actions* 1 .

1. *mvplayer* .

2. *mvplayer* *actions* . Actions

3. 2 . AudioTracks .

CurAudioTrack . 0 .

```
var AudioTracks:Array;
var CurAudioTrack = 0;
```

4. , . audioTracks

AudioTracks . : 'ns.onMetaData'

trace("Subtitle Channels: " + subtitleChannelNumber);

'} ' AudioTracks = infoObject.audioTracks; .

```
ns.onMetaData = function(infoObject:Object)
{
      AudioTracks = infoObject.audioTracks;
}
```

5. *mvplayer* .

'prev_audiotrack' .

6. *mvplayer* actions *Actions* .

prev_audiotrack .

```
prev_audiotrack.onRelease = function()
{
      if (AudioTracks  == undefined || AudioTracks.length < 2)
      {
            return;
      }
      if (CurAudioTrack == 0)
      {
            CurAudioTrack = AudioTracks.length - 1;
      }
      else
      {
            CurAudioTrack--;
      }
      ns.audioTrack = AudioTracks[CurAudioTrack].trackIndex;
}
```

7. 5 *mvplayer*

'next_audiotrack' .

8. *mvplayer* actions *Actions* .

next_audiotrack .

```
next_audiotrack.onRelease = function()
{
     if (AudioTracks  == undefined || AudioTracks.length < 2)
     {
           return;
     }
     CurAudioTrack++;
     if (CurAudioTrack >= AudioTracks.length)
     {
```

```
                CurAudioTrack = 0;
            }
            ns.audioTrack = AudioTracks[CurAudioTrack].trackIndex;
        }
```

9. SWF Scaleform Player .

   Next *Previous* (USM

).

6 액션 스크립트 비디오 확장기능

Scaleform .

Scaleform . NetStream ,

.

6.1 NetStream

`currentFps:Number`

.

: `var currentFps:Number = ns.currentFps;`

`time:Number`

Example: `var currentTime:Number = ns.time;`

6.2 NetStream

`onCuePoint = function(infoObject:Object) {}`

USM , .

주의: 'onCuePoint' .

`onMetaData = function(infoObject:Object) {}`

USM .

주의: 'onMetaData' .

Scaleform Extensions

`audioTracks`

USM

:

```
ns.onMetaData = function(infoObject:Object)
{
      audioTrackArray = infoObject.audioTracks;
}
```

`subtitleTracksNumber`

USM .

:

```
ns.onMetaData = function(infoObject:Object)
{
    numSubtitleChannels = infoObject.subtitleTracksNumber;
     trace("Subtitle Channels: " +  numSubtitleChannels);
}
```

```
onStatus = function(infoObject:Object) {}
```

NetStream .

주의: 'onStatus' .

6.3 NetStream

`close()`

, ns.time 0 .

.

Example: `ns.close();`

`pause([flag:Boolean])`

.

Example: `ns.pause(true);`

`play(name:Object, start:Number, len:Number, reset:Object)`

(USM) .

Example: `ns.play("myVideo.usm");`

`seek(offset:Number)`

( ) .

Example: `ns.seek(50);`

`setBufferTime(bufferTime:Number)`

. GFx::Video::Video

.

.

1 .

: `ns.setBufferTime(2.5);`

6.4 Scaleform

`subtitleTrack = Number`

NetStream

Example: `ns.subtitleTrack = 1;`

`audioTrack = Number`

NetStream .

,

.

: `ns.audioTrack = 2;`

`subAudioTrack = Number`

NetStream SubAudio

.

: `ns.subAudioTrack = 3;`

`voiceTrack = Number`

5.1 (center track) . 5.1

.

: `ns.voiceTrack = 1;`

`setReloadThresholdTime(reloadTime:Number)`

.

. 4 (setBufferTime)

1 , GFx::Video::Video

4 . 3 ,

GFx::Video::Video .

: `ns.setReloadThresholdTime(1);`

`setNumberOfFramePools(numPools:Number)`

. GFx::Video::Video

. CPU

. 2 .

: `ns.setNumberOfFramePools(3);`

`setOpenTimeout(timeout:Number)`

open file timeout(USM) .

5 . (timeout) 0 .

: `ns.setOpenTimeout(5);`

currentFrame:Number

( ) .

: trace("Current FPS: " + ns.currentFps + " fps");

Loop = Boolean

NetStream loop true false .

Example: ns.loop = true;

clipRect = new rectangle(x:Number, y:Number, width:Number, height:Number)

.

Example: ns.clipRect = new rectangle(100, 100, 240, 200);

onSubtitle = function(msg:String) {}

USM .

Example: ns.onSubtitle = function(msg:String) { trace(msg); }

7 비디오 생성에 대한 이해

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. ,

.

, , ,

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. , .

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. ,

, , , ,

.

HD 1080p XB360 PS3 ,

. ,

HD

. HD .

HD , 1080p

.

.

. .

, .

.

AVI .

. (DIVX)

,

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. , , 5.1

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,

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. ,

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.

, ,

.

7.1

h□9 ■□

8 기술적 통합 지침

.

8.1 Scaleform

Scaleform GFx::Audio GFx::Loader

. SWF

SoundRenderer . SoundRenderer

C++ . Scaleform

FMOD .

Sound::SoundRendererFMOD::CreateSoundRenderer FMOD::System

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사용예:

```
// Initializing the sound renderer object
   FMOD::System* pFMOD = NULL;
   result = FMOD::System_Create(&pFMOD);
   if (result != FMOD_OK)
      exit(1);
   result = pFMOD->getVersion(&version);
   if (result != FMOD_OK || version < FMOD_VERSION)
      exit(1);
   result = pFMOD->init(64, FMOD_INIT_NORMAL, 0);
   if (result != FMOD_OK)
      exit(1);
   Ptr<SoundRenderer> psoundRenderer =
               *SoundRendererFMOD::CreateSoundRenderer();
   if (!psoundRenderer->Initialize(pFMOD))
      exit(1);

   // Setting an instance of GFx::Audio on the loader
   Ptr<GFx::Audio> paudioState = *new GFx::Audio(psoundRenderer);
   loader.SetAudio (paudioState);
```

8.2 Scaleform

Scaleform Video GFx::Loader

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VideoSoundSystem GFx::Video::Video

. Scaleform SoundRenderer .

사용예:

```
 // Setting an instance of Video on the loader. An optional parameter
//  allows to specify the priority of the video decoding thread.
Ptr<GFx::Video::Video> pvc = *new Video(Thread::NormalPriority);
// Setting a video sound system instance which is based on SoundRenderer
//interface
if (psoundRenderer)
{
      pvc->SetSoundSystem(psoundRenderer);
}
// Setting a video sound system instance which is specific to a particular
//platform.
// Note: SetSoundSystem method should be called only once per an instance of
Video
 pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new
                              VideoSoundSystemDX8(0)));

loader.SetVideo(pvc);
```

Video::Video VideoPC 3 .

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, (affinity) .

```
VideoPC(Thread::ThreadPriority
          decodingThreadsPriority = Thread::NormalPriority,
          int decodingThreadsNumber = MAX_VIDEO_DECODING_THREADS,
          UInt32* affinityMask = NULL);
```

사용예:

```
UInt32 affinities[] = {
                    DEFAULT_VIDEO_DECODING_AFFINITY_MASK,
                    DEFAULT_VIDEO_DECODING_AFFINITY_MASK,
                    DEFAULT_VIDEO_DECODING_AFFINITY_MASK };

Ptr<Video::Video> pvc = *new Video::VideoPC(Thread::NormalPriority,
                                MAX_VIDEO_DECODING_THREADS, affinities);
```

PS3 XBox360 Scaleform Video ,

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PS3 GFx::Video::VideoPS3 5 .

```
VideoPS3(CellSpurs* spurs,
          int ppu_num, Thread::ThreadPriority ppu_priority,
          int spu_num, UInt8* spurs_priorities = NULL);
```

spurs - CellSpurs

spu_num - SPU .

ppu_num - PPU

ppu_priority - PPU

spurs_priorities- SPU

사용예:

```
UInt8 spurs_priorities[] = {1,1,1,1,0,0,0,0};
Ptr<Video::Video> pvc = *new VideoPS3(GetSpurs(), 0, 1000,
                                        4, spurs_priorities);
```

PPU 사용률 최소화 예제:

Scaleform , PPU . , 720p

PPU 5% 50~60FPS .

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PPU ,

PPU 40% . ,

, ,

4% 5% .

PPU 1% .

```
UInt8 spurs_priors[] = {0,0,1,1,1,1,0,0};
Ptr<Video::Video> pVideo = *new VideoPS3(GetSpurs(), 0,
                        Thread::BelowNormalPriority, 4, spurs_priors);
```

XBox360 VideoXbox360

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```
VideoXbox360(unsigned decodingProcs = Xbox360_Proc1 | Xbox360_Proc3 |
                        Xbox360_Proc5, Thread::ThreadPriority
                        decodingThreadsPriority = Thread::NormalPriority,
                        Xbox360Proc readerProc = Xbox360_Proc2);
```

사용예:

```
      Ptr<Video::Video> pvc = *new VideoXbox360(Xbox360_Proc1 |
                                                Xbox360_Proc3|Xbox360_Proc5,
                                                Thread::NormalPriority,
                                                Xbox360_Proc2);
```

8.3 API

Video::VideoBase::ReadCallBack

VideoBase::IsIORequired, VideoBase::EnableIO .

ReadCallback

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예:

```
//
class VideoReadCallback : public VideoBase::ReadCallback
{
   public:
   VideoReadCallback() {}
   virtual ~VideoReadCallback() {}

   virtual void OnReadRequested()
   {
      //
      VideoReadStart = Timer::GetTicks();
      // ...
   }
   virtual void OnReadCompleted()
   {
      //
      GameReadStart = Timer::GetTicks();
      // ...
   }

   UInt64  VideoReadStart;
   UInt64  GameReadStart;
};

//
Ptr<Video::Video> pvc = *new Video::VideoPC(
         Thread::NormalPriority, MAX_VIDEO_DECODING_THREADS, affnts);

//
Ptr<VideoReadCallback> pvideoReadCallback = *new VideoReadCallback;
pvc->SetReadCallback(pvideoReadCallback);
```

IsIORequired

. true IO

EnableIO .

EnableIO /

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예:

```
if(pvc->IsIORequired())
{
    pDataLoader->Enable(false);
    pvc->EnableIO(true);
    fprintf(stderr, "Video is loading\n");
}

// ...

if(!pvc->IsIORequired())
{
    pvc->EnableIO(false);
    pDataLoader->Enable(true);
    fprintf(stderr, "Game data is loading\n");
}
```

(VideoBGLoadFlash

VideoBGLoadData) *Apps\Samples\Video* .

- VideoBGLoadFlash –

  .

- VideoBGLoadData –

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8.4 Scaleform VideoSoundSystem

GFx::Video::VideoSoundSystem Video::Video

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Video::SetSoundSystem()

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Video .

- Video::VideoSoundSystemDX8 – Windows DirectSound
- Video::VideoSoundSystemXA2 – Windows Xbox360 XAudio2
- Video::VideoSoundSystemPS3 – PS3 MultiStream
- Video::VideoSoundSystemWii – Wii
- Video::VideoSoundSystemFMOD - FMOD
- Video::VideoSoundSystemWwise – Wwise

Scaleform

, . ,

Scaleform Sound::SoundRenderer

Video::VideoSoundSystem . (8.1)

VideoSoundSystem VideoSound ,

SoundRenderer . SoundRenderer

, Video

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```
#include "Video\Video_VideoSoundSystemXA2.h"
pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemXA2(0, 0)));
```

GFx::Video::VideoSoundSystem GFx::Video::VideoSound

. VideoSoundSystem

. VideoSoundSystem VideoSound

, Create . Scaleform

. (.) VideoSound

, Scaleform

. VideoSound::PCMStream

. VideoSoundSystem

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VideoSoundSystemWwise Wwise SDK v2009.2.1 build 3271

. Audiokinetic

/ ,

S*DK\samples\Plugins\AkAudioInput* . Wwise

. GFx::Video::Video Wwise SDK

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:

```
#include "Video\Video_VideoSoundSystemWwise."
Ptr<Video::VideoSoundSystem> wwise =
                    Ptr<Video::VideoSoundSystem>(*new  VideoSoundSystemWwise());
pvc->SetSoundSystem(wwise);
wwise->Update();
```

Apps\Samples\FxPlayer and *Apps\Samples\Common* Scaleform Player –

FxPlayerAppBase.cpp, FxSoundFMOD.cpp, FxSoundWwise.cpp ,

Src\Video

Video_VideoSoundSystem*.cpp .

8.5

. GFx::Video::Video

ActionScript

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NetStream audioTrack .

: `ns.audioTrack = 2;`

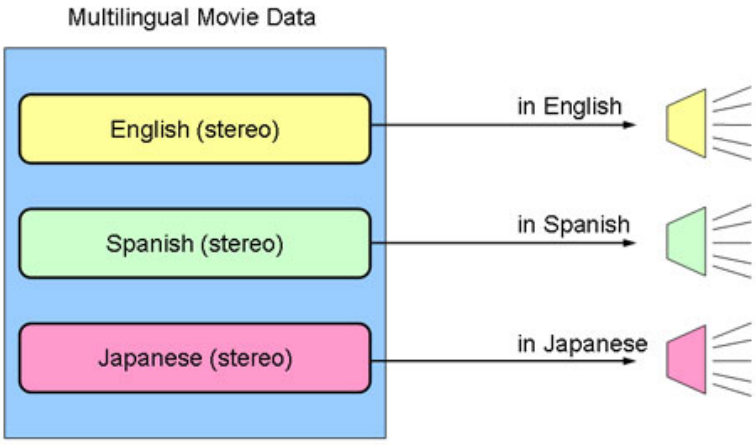

그림 12: 다중 언어 비디오

5.1 .

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GFx::Video::Video .

1. (Center Channel Replacement)
2. SubAudio

8.5.1 중앙 채널 교체(Center Channel Replacement)

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(monaural) . (monaural) 5.1

(center track) .

5.1 (monaural) .

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:
```
ns.audioTrack = 0;   // main 5.1 audio track
ns.voiceTrack = 5;   // mono audio track
```

5.1 (monaural)

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.

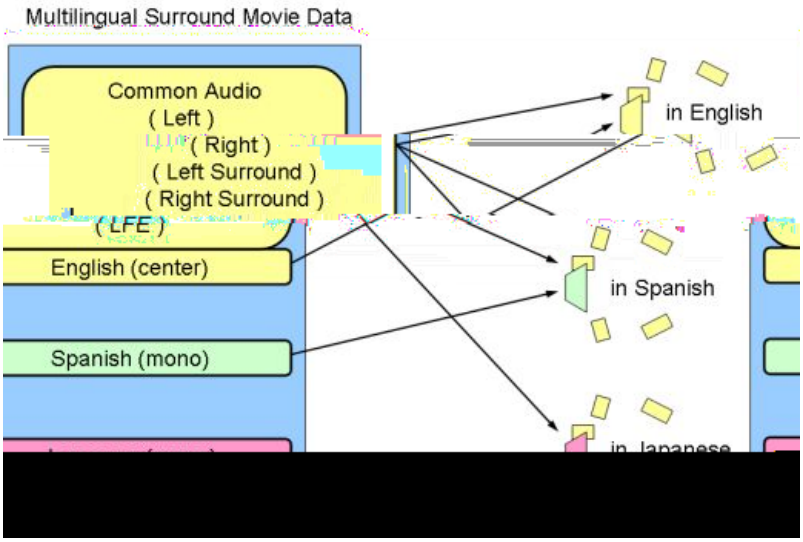

그림 13: 중앙 채널 교체(Center Channel Replacement)

8.5.2 SubAudio 재생

SubAudio GFx::Video::Video

. , GFx::Video::Video

. , SubAudio

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:

```
ns.audioTrack = 0;       // main audio track
ns.subAudioTrack = 6;    // secondary audio track
```

SubAudio

. Sound.setVolume() .

:

```
var audio:Sound = new Sound(this);
audio.setVolume(80, 50);       // set volume for main and subaudio tracks
```

그림 14: SubAudio 재생

9 문제 해결

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1. (, IO
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2. .

3. (,)

, ( ).

1. – .
   . (720P)
   3 4 .
   8 .

2. – /
   . Scaleform
   .

3. – ActionScript
   .
   . 6 .
   a. `NetStream.setNumberOfFramePools(numPools:Number)`
   b. `NetStream.setBufferTime(bufferTime:Number)`